Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week-1

2520030078

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Section - 3

Query 1: Create the database

```
1 •  CREATE DATABASE IF NOT EXISTS bookflows_db;
```

Output:

```
1  20:24:43  CREATE DATABASE IF NOT EXISTS bookflows_db
```

Query 2: Select the database

```
2 •  USE bookflows_db;
```

Output:

```
2  20:26:17  USE bookflows_db
```

Query 3: Create the books table

```
1 •  CREATE DATABASE IF NOT EXISTS bookflows_db;
2 •  USE bookflows_db;
3 •  CREATE TABLE books (
4        book_id INT AUTO_INCREMENT PRIMARY KEY,
5        title VARCHAR(255) NOT NULL,
6        isbn VARCHAR(13) NOT NULL UNIQUE,
7        published_year INT,
8        CONSTRAINT chk_published_year CHECK (published_year < 2027)
9    );
10 •  SELECT * FROM books;
```

Output:

```
3  20:28:11  CREATE TABLE books ( book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL, i...  0 row(s) affected
```

book_id	title	isbn	published_year
NULL	NULL	NULL	NULL

Query 4: Create the members table:

```
1  CREATE DATABASE IF NOT EXISTS bookflows_db;
2  USE bookflows_db;
3  CREATE TABLE books (
4   book_id INT AUTO_INCREMENT PRIMARY KEY,
5   title VARCHAR(255) NOT NULL,
6   isbn VARCHAR(13) NOT NULL UNIQUE,
7   published_year INT,
8   CONSTRAINT chk_published_year CHECK (published_year < 2027)
9  );
10 SELECT * FROM books;
11 CREATE TABLE members (
12  member_id INT AUTO_INCREMENT PRIMARY KEY,
13  full_name VARCHAR(100) NOT NULL,
14  email VARCHAR(150) NOT NULL UNIQUE
15 );
16 SELECT * FROM members;
```

Output:

member_id	full_name	email
NULL	NULL	NULL

Query 5 (verification): Inspect the table structures

```
DESCRIBE books;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(255)	NO		NULL	
isbn	varchar(13)	NO	UNI	NULL	
published_year	int	YES		NULL	

```
18 DESCRIBE members;
```

Output:

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

Query 6: Insert 3 books

```
INSERT INTO books (title, isbn, published_year) VALUES
('The Alchemist', '9780061122415', 1988),
('Clean Code', '9780132350884', 2008),
('Atomic Habits', '9780735211292', 2018);
```

Output:

```
10  20:42:08  INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Cle
```

Query 7: Insert 3 members

```
INSERT INTO members (full_name, email) VALUES
('Gutha Sushma Sree', 'sush@example.com'),
('Gutha Sowmya Sree', 'sow@example.com'),
('Gutha Mokshagna Naidu', 'moksha@example.com');
```

Output:

```
14  20:44:40  INSERT INTO members (full_name, email) VALUES ('Gutha Sushma Sree', 'sush@example.com'), ('Gutha Sov
```

```
SELECT * FROM books;
```

Output:

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018
NULL	NULL	NULL	NULL

```
SELECT * FROM members;
```

Output:

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com

Constraint Tests (These MUST Fail):

Test 1: Duplicate ISBN (UNIQUE violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Fake Copy', '9780061122415', 2000);
```

Output:

```
26  09:10:01  INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000)
```

Test 2: NULL title (NOT NULL violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES (NULL, '9999999999999', 2010);
```

Output:



Test 3: Future publication year (CHECK violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Time Traveler', '8888888888888', 2030);
```

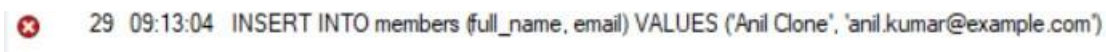
Output:



Test 4: Duplicate email (UNIQUE violation):

```
INSERT INTO members (full_name, email)
VALUES ('Anil Clone', 'anil.kumar@example.com');
```

Output:



----------X----------